Dynamic Interactive Learning for Lightweight Detectors in Remote Sensing Imagery

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***Abstract*— The lightweight model has played an important role in the remote sensing (RS) realm. The existing researchers have proposed many models with lightweight structures, but their performance still has a gap compared with the deep model. A promising approach to optimize the lightweight model is knowledge distillation (KD), which can be viewed as knowledge transfer from the teacher model. However, the existing KD approaches have some issues. On one hand, offline distillation methods usually ignore the interactive learning between the student model and the teacher model. On the other hand, knowledge transfer does not consider instance property. This offline distillation strategy without property perception may not suitable for multiscale, diverse, and complex RS instances and results in a suboptimum training status. In this article, we propose a dynamic interactive learning (DIL) framework for optimizing RS lightweight detectors. First, we propose an instance interaction learning module. It calculates the value of every instance in the batch of the teacher and student prediction by each model's real-time state and instance property. Then according to the DIL thought, we facilitate the low-quality instance to learn from the high-quality one whether it is from the teacher or student model. Moreover, we also propose the instance property perception (IPP) strategy that weighs the distillation knowledge of instances according to their feature, category, and location property. In the proposed DIL framework, both the teacher and student models are trained together and it is cost-free in the testing phase. Extensive experiments on three RS datasets demonstrate the effectiveness of the DIL.**

***Index Terms*— Knowledge distillation, lightweight model, object detection, remote sensing (RS) imagery.**

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NOMENCLATURE

S	Student model.
T	Teacher model.
θ_s	Student model parameters.
θ_t	Teacher model parameters.
Φ	Knowledge distillation process.
I	Instance set.
P	Instance property.

I. INTRODUCTION

THE odyssey of neural network design is on a path to become progressively deeper in recent years and it acquires more and more gains. But we should ask ourselves a question, do deep models really can be used in mobile and embedded equipment that cannot afford high computational needs? With the development of aerial technology, some lightweight devices, such as drones and satellites, are faced with the problem of limited resources which cannot deploy the deep model. Motivated by the issue, more and more remote sensing (RS) researchers have focused on the lightweight model realm.

To simplify the complicated model structure, recent researchers have proposed some lightweight models, such as MobileNet [1], ShuffleNet [2], and SqueezeNet [3]. However, a bad performance is usually displayed when directly adopting them as the backbone of RS downstream tasks. The low capacity of the backbone results in a weak feature extraction ability which leads to worse classification and location for complex RS objects. A perceptual intuition is that some approaches need to be used to facilitate the lightweight model to get out of the suboptimum solution, which would be better for the object detection task. Recently, knowledge distillation (KD) has been verified as an efficacious approach for optimizing the lightweight model, which introduces some extra dark knowledge from another well and pretrained teacher model output. Learning efficient object detection models with knowledge distillation (LEOD) [4] first suggests adopting the KD in the detection task, which transfers various knowledge from the teacher model to the student model. Distilling object detectors with fine-grained feature imitation (FGFI) [5] suggests the student should imitate the positive region of interest (RoI) regions which are close to the ground truth. Distilling object detectors with task adaptive regularization (TAR) [6] puts forward that the ground-truth regions are more valuable

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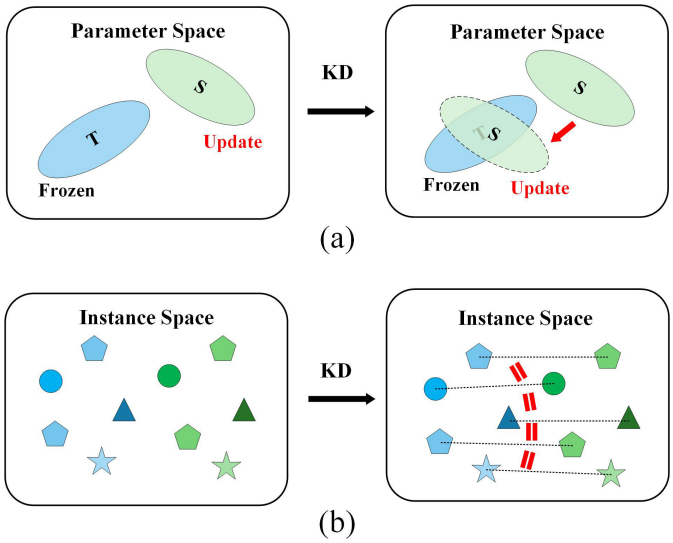


Fig. 1. Issues of the existing approach. (a) KD process is viewed as minimizing the parameter distribution gap between the teacher model and the student model, but only updating the student may not validly achieve the KD goal. (b) In the task of object detection KD, each instance usually is the core of learning. But assigning equivalent learning weight without considering instance property can make distillation lack uniqueness and thus yield suboptimal results.

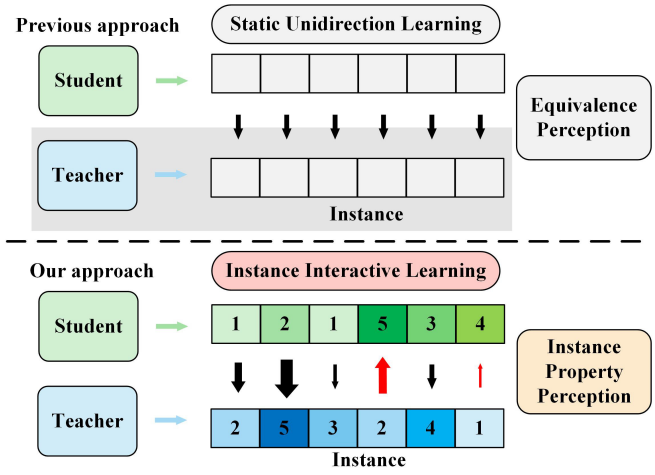


Fig. 2. Comparison of the task framework. The previous approaches do not consider the instance property and the student unidirectionally learns from the teacher model. Our method is instance-level interactive learning and transfers the knowledge according to the instance property.

to learn. However, the above methods still have some issues which are not better optimizations for the RS lightweight model. Two main reasons below cause the suboptimum results.

A. Offline Unidirectional Distillation

The previous methods usually focus more on offline unidirectional distillation, which adopts a well-pretrained teacher model as the guidance. The parameters of the teacher model are frozen during training and only the student model is updated [see Fig. 1(a)]. Obviously, there is a training gap between the teacher model and the student model, which includes the network gap and input data gap. In specific, the network gap is owing to backbone capacity. The input data gap is caused by the randomness in mini-batch samples of the teacher and student's respective training phases, which is an asynchronous process. Besides, RS imagery is captured by different satellites and aircraft, which means that data gaps are greater compared with natural images. The existing offline distillation methods magnify the RS data asynchronous process and lead to suboptimal KD in RS lightweight detector. Besides, as an old saying goes, "Nobody is perfect," the teacher is not correct in every instance prediction compared with the student. The student even can teach the teacher knowledge in some instances' prediction. In addition, the teacher model is unable to synchronous collaborative learning with the student model owing to the frozen parameters. Hence, there still exists a greater improvement for offline unidirectional distillation.

B. No-Perception for Instance Property

The second issue is produced by the essence of the detection task, which includes classification and regression by instance features. In general, the feature, category, and location are considered as three core information of the instance. Previous approaches usually ignore the instance property and distill instances equally [see Fig. 1(b)]. However, objects in RS imagery usually have diverse sizes, complex features, and irregular shapes compared with natural objects. The previous distillation strategies do not consider the property of the instance. Objects of extreme and normal size, big, and small size are usually given the same distillation weight. It seems not suitable for varied RS instances distillation. Hence, a dynamic weight should be deployed to different instances according to its three properties for better distillation.

Driven by the above issues, we propose a dynamic interactive learning (DIL) framework, which involves colearning of the teacher model and the student model and guiding each other with instance property perception (IPP). As illustrated in Fig. 2, the previous approaches usually let the student model learn each instance knowledge from the teacher model regardless of the instance property. However, our method is different from theirs. First, the proposed instance interaction learning (IIL) module can dynamically calculate each instance value of the teacher model and the student model according to their real-time state and instance prediction. Besides, to better transfer knowledge in an interactive way instead of the previous teacher guide mode, we facilitate the low-quality instance to study from the high-quality instance. This instance-level DIL makes the teacher and student models help each other and find the optimal model parameters. Moreover, we also design an IPP strategy, which weighs the instance distillation knowledge according to its feature, category, and location. Through the IPP strategy including feature enhancement, category balance, and location calibration, our lightweight model can better learn the knowledge of hard instances and detect them.

Our main contributions and innovations are summarized as follows.

1) We propose a general DIL framework to optimize the lightweight detectors in the RS imagery. Extensive experiments on three benchmarks DOTA, NWPU VHR-10, and DIOR demonstrate the effectiveness of our approach, which transcends other state-of-the-art (SOTA) KD approaches. Moreover, the accuracy of the

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lightweight model optimized by DIL surpasses other heavy or lightweight detectors.

2) The proposed instance interaction learning module measures the value of each instance from the teacher model and the student model. The instance values are calculated by their property and the real-time state of the model. We facilitate low-quality instances to learn from the high-quality ones no matter whether it is from the teacher or student. DIL keeps the training consistent between the student model and the teacher model, which is better for KD.
3) To tackle the issue of equal distillation for diverse RS instances in previous methods, we design an IPP strategy, which considers the properties including feature, classification, and regression. We adopt the strategy to weigh the knowledge of each instance for adaptive distillation.

We schedule the remaining article as follows. Section II introduces the related works. Our method is presented in detail in Section III. The experiments are shown and analyzed in Section IV. Finally, we summarize our work in Section V.

II. Related Works

In this part, we introduce the related works which include object detection, lightweight model, and KD.

A. Object Detection

With the rapid development of hardware equipment and deep learning, the accuracy and speed of object detection have been greatly improved. Currently, the convolutional neural network (CNN)-based object detection algorithms are divided into two schools, which include two-stage and one-stage detectors. In general, two-stage detectors such as Faster R-CNN [7] propose lots of proposals in the first stage, and then regress and classify them in the second stage. Although Faster R-CNN [7] has a good detection result, the detection speed is relatively slow and cannot realize real-time detection. YOLO [8] as a one-stage detector directly predicts the position and category of the object from the feature map and achieves a higher speed. Subsequent one-stage detectors are proposed to improve accuracy, such as YOLOv2 [9], YOLOv3 [10], and single shot multibox detector (SSD) [11]. But there is still an existing accuracy gap compared with the two-stage detectors. To reduce the performance gap, some new techniques are proposed for the one-stage detectors. In RetinaNet [12], the focal loss is used to replace the original cross-entropy function, which suppresses the impact of the imbalance of samples. ATSS [13] proposes the adaptive training sample selection algorithm that obtains an adaptive intersection over union (IoU) threshold by the statistical information of IoU at each level to select a close amount of positive and negative samples for each object.

However, multiscale objects and complex backgrounds in the huge-scale RS imagery of the above detectors cannot satisfy the detection task. Therefore, many researchers have proposed the RS-character-based methods for RS imagery. Wang et al. [14] use the atrous spatial feature pyramid (ASFP) module to integrate multiscale features and propose the area-weight function to pay attention (AT) to objects with small sizes. Sun et al. [15] put forward the context-based deformable module (CBD) module to enhance the detection results of rigid objects by combining deformable convolution network (DCN) and attention mechanism. Wei et al. [16] detect the aircraft by clustering the intersection lines paired in airplanes. Besides, the new pentagonal mask proposed by them can provide a more precise location. Zhu et al. [17] propose an automatic organization mechanism to detect rotating objects.

B. Lightweight Model

The classic detectors [7], [8], [11], [12], [13] are usually complex and consume massive computation, which is not suitable for lightweight detection tasks. The heavy backbones including VGG16 [18] and ResNet [19] are the major reason for their cumbersome detectors. To simplify complicated structure, some backbone designs for the lightweight model such as MobileNet [20] and ShuffleNet [2] are proposed. Furthermore, lightweight detectors such as MobileNet-SSD [20], MobileNetV2-SSDLite [1] are designed when MobileNet [20] is combined with the single-stage detectors.

Although MobileNet [20] plays an important role in the lightweight model, the original design is for the classification task. Adopting it as the backbone for downstream detection tasks may lead to a suboptimum solution. Therefore, Pelee [21] and Tiny-DSOD [22] aim to design a special lightweight backbones for detection task. Besides, ThunderNet [23] has the same thought, which proposes to devise the compressed SNet, region proposal network (RPN), and task heads. Moreover, ThunderNet can detect real-time on advanced RISC machine (ARM) platforms finally.

C. Knowledge Distillation

KD can be viewed as complementary learning and the student learns dark knowledge from the teacher model except for the ground truth, which optimizes the lightweight model from another perspective. Recently, various approaches are proposed and we introduce them from the classification task to the detection task.

KD has been first proposed by Hinton et al. [24] in the classification task. They consider the soft logit outputs from the teacher are useful for student model optimization, which let the student learn the potential relationship in the data. Since then, more and more works are paid attention to this realm. FitNet [25] proposes the intermediate layers named "hint" that can transfer feature knowledge to the student model. The "hint learning" makes the student imitate the feature information of the teacher model. AT [26] designs a complex attention mechanism to deliver some more valuable information. Distilling knowledge via knowledge review (DKKR) [27] proposes a knowledge review mechanism by the cross-stage connection paths, and the experiment results reveal its great importance.

Research of KD in classification tasks focuses on which type of knowledge is valuable for distillation. However, the detection distillation task is more complex owing to the imbalance between the objects and the background. More works devote themself to the selection of object knowledge instead of all knowledge of images. LEOD [4] first adopts

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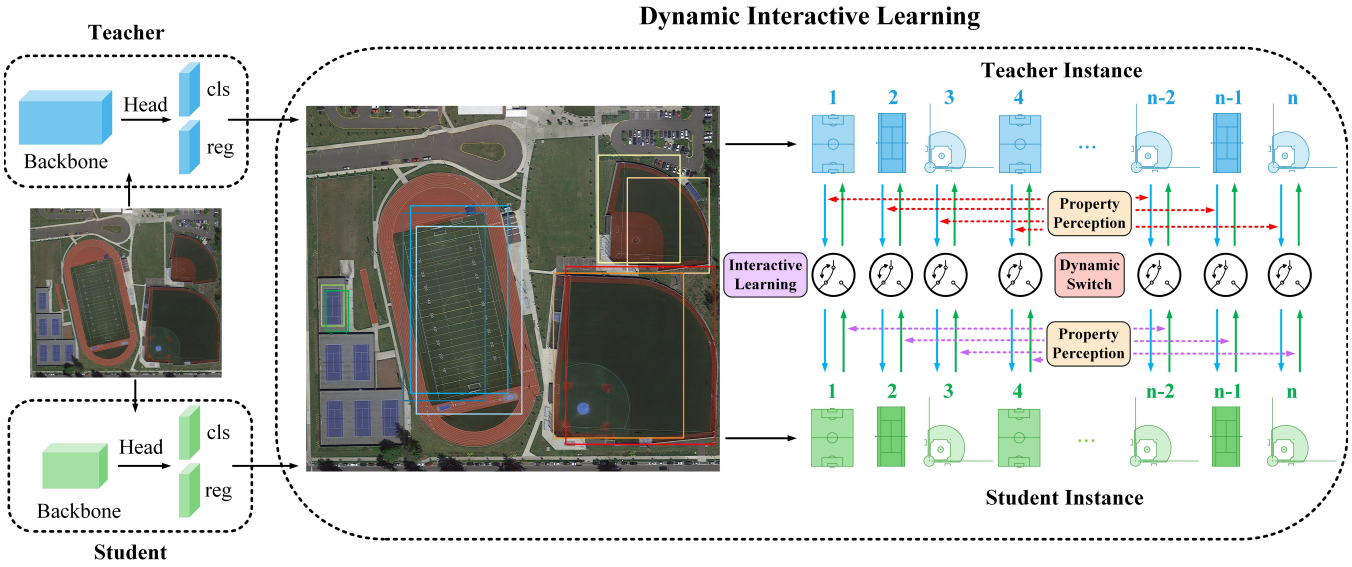


Fig. 3. Pipeline of the DIL framework. First, we get the corresponding instance prediction from the teacher and the student, respectively. Then, values of all the instances are calculated and a dynamic switch is adjusted by the comparison of instance values. Hence, the low-quality instance will learn from the high-quality one. Moreover, IPP will weigh the transferring knowledge according to instance prediction.

the KD thought to optimize the student model in the detection task. Different from the classification, they transfer the features and prediction results to the student model which improves performance a lot. Mimick [28] points out that RoI regions are more valuable for learning, which contains positive and negative samples. FGFI [5] suggests that the student learns the fine-grained feature of the teacher model which is better for detection. TAR [6] proposes a Gaussian mask to enhance object information and suppress those background regions. They also design an adaptive distillation decay to better converge the model. DeFeat [29] suggests that both background feature and object feature should be distilled together but they need to be assigned different weights. Localization distillation for dense object detection (LD) [30] proposes to view the regression knowledge as location probability distillation, which shows more efficiency than semantic knowledge for distilling object detectors. Adaptive reinforcement supervision distillation (ARSD) [31] proposes to adaptively select the core features from multiscale levels and choose the high-quality box location for distillation. Statistical sample selection and multivariate knowledge mining for lightweight detectors in remote sensing imagery (S^3MKM) [32] suggests choosing statistical samples for distillation and transferring their explicit and implicit knowledge.

III. Methods

The whole framework of DIL is illustrated in Fig. 3. We first introduce the DIL framework in Section III-A. Then instance interactive learning (IIL) and IPP are presented in Sections III-B and III-C, respectively. We finally elaborate on the optimization in Section III-D.

A. Summarization

KD is an effective approach that delivers extra information (features, soft classification, and location knowledge) from the teacher model to the student model. The knowledge transmission helps the student optimize the model parameters. Compared with the classification task that usually uses the whole image as a knowledge base, the complicated detection task will focus on the instance-level KD.

Before introducing the KD, we first give some definitions of the instance. Samples are acquired by model forward propagation and instances are screened out from samples according to their relationship with ground truth as below

$$I = M(\mathcal{I}, \theta_m, \psi) \tag{1}$$

where $M, \mathcal{I}, \theta_m,$ and ψ denote the model, image, network parameters, and threshold, respectively. For a better description, we can also define the instance set as follows:

$$I = \{i_1, i_2, \ldots, i_n\} \tag{2}$$

where n represents the number of instances in an image. In general, instances are positive samples and they all have three properties as below

$$P = \{f, y, b\} \tag{3}$$

where $f, y,$ and b denote the feature, classification, and regression of instance, respectively. Not only does the property P represent knowledge but also it reveals the potential value of knowledge, and we will introduce it in Section III-C in detail.

Nomenclature sums up some related definitions for a more clear description. In the KD process, we optimize θ_s to close to θ_t by transferring the dark knowledge. Hence, the overall process is presented below

$$\Phi(\theta_s) = \arg\min_{\theta_s} \mathrm{KL}[\theta_t || \theta_s] \tag{4}$$

where KL means the Kullback–Leibler divergence. It may be difficult to directly reduce the difference in the distribution

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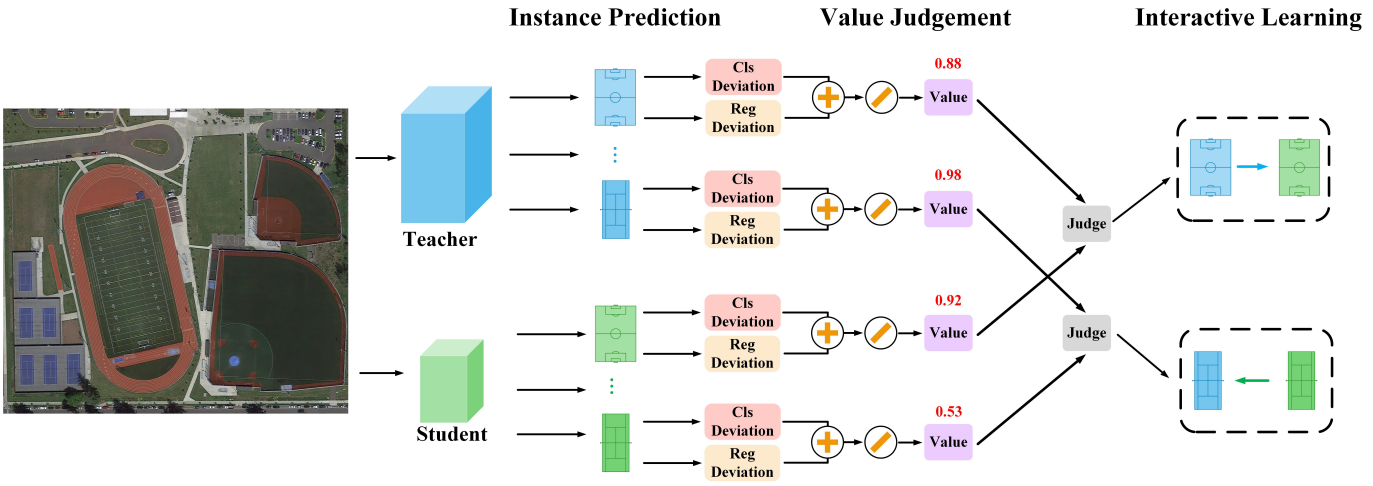


Fig. 4. Illustration of the IIL module. For the same input image, the teacher model and student model will output the corresponding instances. To get the instance value, we calculate the classification and regression deviation of the instance and obtain the value by (8). In the second stage, we compare the instance value from the teacher model and the student model and facilitate low-quality instances to learn from high-quality ones, which is a dynamic interactive process.

between θ_s and θ_t, so we optimize θ_s by imitating the instance property of the teacher model. So (4) is transformed as follows:

$$\Phi(\theta_s) = \arg\min_{\theta_s} \sum_{i\in I}\sum_{j\in P} \omega_{ij}\mathcal{L}_j\left(S_i^j, T_i^j\right) \tag{5}$$

where $\mathcal{L}$ represents the loss function. We adopt different loss functions for different properties. Moreover, ω is the dynamic weight for each property of the instance.

Based on the above theory, we propose a DIL framework for lightweight RS detector optimization. Fig. 3 provides a diagram of the DIL framework, which includes IIL and IPP two modules. The IIL module let the corresponding instance of the teacher model and the student model learn from each other according to their value comparison. The IPP module weighs the distillation knowledge by the instance property. We elaborate on them in detail in Section III-B.

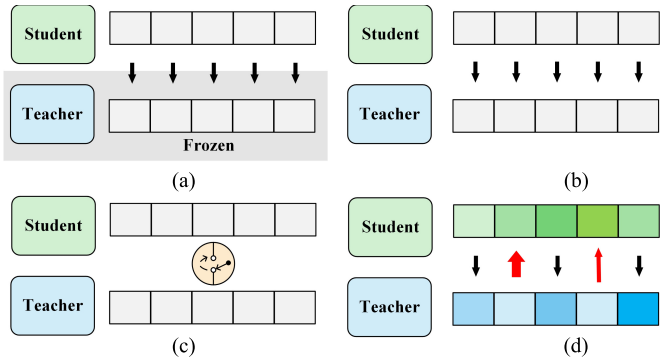


Fig. 5. Illustration of the four distillation frameworks. The arrows in (a) and (b) are static knowledge learning, and the arrows in (d) are dynamic knowledge learning. (a) Classic distillation strategy and the frozen teacher teaches the updated student. (b) Both the teacher and the student are updated. (c) Student and teacher roles are determined by their state. (d) Our instance-level interactive learning.

B. Instance Interaction Learning

Previous approaches treat KD as a decoupled process and distill key components of the detector, respectively. In general, the backbone, classification head, and regression head contain valid information for distillation. In the proposed theory, we focus more on knowledge carried by instance and regard distillation as instance-level information delivery.

As defined in (2), we first obtain the instance set I for an input image by a standard detector label assignment process. Both the student and teacher models are single-stage models in our distillation pipeline, and hence, the set I_t and I_s are alignments

$$I \sim I_t \sim I_s. \tag{6}$$

Different from the traditional perspectives, we propose the instance property [see (3)] as key information for distillation. The instance property has two functions, one is to evaluate the instance value, and the other is to transfer knowledge. In this section, we mainly adopt the value evaluation function to realize instance interaction learning.

According to the label assignment process, each instance has a ground truth as the target to learn. We can calculate the deviation between the instance and the ground truth. The fewer the deviation, the bigger the instance value, which means the instance predicts a more accurate object. We calculate the deviation as follows:

$$\begin{aligned} D(i) &= \mathcal{L}(i, \mathcal{Y}^i) \\ &= \mathcal{L}_{\text{cls}}(y_i, \mathcal{Y}^i) + \mathcal{L}_{\text{reg}}(b_i, \mathcal{Y}^i) \end{aligned} \tag{7}$$

where $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{reg}}$ represent the classification and regression deviation, respectively. In general, we adopt the focal loss [12] to evaluate the classification deviation and use the generalized intersection over union (GIoU) loss [33] to measure the regression deviation. The value is obtained below

$$V(i) = 1/D(i). \tag{8}$$

Loop (7) and (8) for the set I_t and I_s, we acquire V_t and V_s, respectively.

Before diving into the instance interaction learning presented in Fig. 4, we show some existing distillation strategies in Fig. 5. To be specific, Fig. 5(a) presents the basic distillation strategy. Only the teacher model is frozen and the student learns all the knowledge from the teacher without value judgment. Fig. 5(b) is the upgrade of Fig. 5(a) because the teacher

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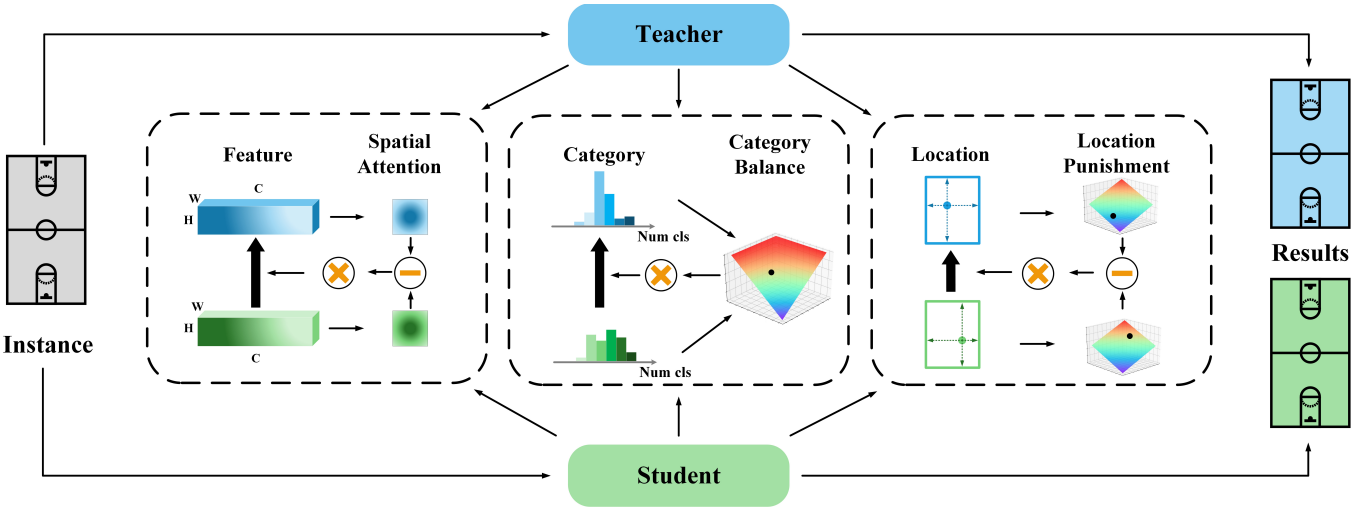


Fig. 6. Illustration of the IPP module. The IPP contains three perception strategies including feature enhancement, category balance, and location calibration. First, the teacher and student models predict the feature, category, and location for the same instance, respectively. Then, the IPP module dynamically weighs the distillation knowledge transferred from the teacher model to the student model. In general, the feature enhancement strategy adopts the spatial attention weight to shrink feature differences between the teacher model and the student model. The category balance strategy guides the student's instance of harder and fewer category objects to learn more knowledge from the teacher model. The location calibration strategy helps the student learn more accurate location of bounding boxes from the teacher model.

model is also updated during training for synchronization. Fig. 5(c) has improved the knowledge delivery way compared with Fig. 5(b). The role of the teacher is dynamic, which is determined by the statistical information comparison of the mini-batch instance of a big or small model. Fig. 5(d) is our IIL that is more interesting; we realize the instance-level role decision. But in Fig. 5(c), the role of the teacher is restricted by mini-batch data. The mini-batch samples are all the teachers or students instead of independent instances. In conclusion, our IIL is the upgrade of Fig. 5(a)–(c). First, the teacher model is also updated during training, which can reduce the data gap between the student and the teacher. Second, our strategy considers that each instance has independent distillation and knowledge flow is dynamic. Therefore, the teacher and the student learn from each other at the instance level.

To realize instance interaction learning, we compare the corresponding instance value between the teacher and the student. We facilitate the low-quality instance to learn from the high-quality one no matter whether it comes from the teacher or student. We define the learning process as follows:

$$\begin{aligned} T(i) &\overset{\omega_i}{\Leftarrow} S(i), \quad \text{if } V(i^s) < V(i^t) \\ S(i) &\overset{\omega_i}{\Leftarrow} T(i), \quad \text{if } V(i^s) > V(i^t) \end{aligned} \tag{9}$$

where $\Leftarrow$ represents the learning process. Specifically, the student learns from the teacher model when the teacher instance value is greater than the student. Otherwise, the teacher studies the student. Besides, ω denotes the distillation weight brought by instance property gain, which is introduced in detail in Section III-C.

C. Instance Property Perception

Instances represent potential objects in the image, and hence adopting instances for distillation can reduce the impact of complex backgrounds. Besides, the knowledge carried by instances is the key information for distillation. We consider three properties $P = \{f, y, b\}$ as defined in (3). To be specific, f is the feature of the instance, y denotes the classification predictions of the instance, and b represents the regression results of the instance. Three properties will be introduced in the perception strategies below, respectively.

1) Feature Enhancement: Feature maps are the core information of the instance, and the subsequent classification and location are done based on it. Define the feature maps of the instance i as $f_i \in R^{1\times1\times C}$. An activation-based mapping conversion F make the 3-D tensor f_i transform to a spatial attention weight

$$F : R^{1\times1\times C} \to R^1. \tag{10}$$

Before defining such a spatial attention conversion, we make an important assumption that the absolute value of an instance feature can be adopted as an indication of the importance of the instance. Therefore, we can construct the instance spatial attention weight by computing statistics of these values across the channel dimension. Specifically, two activation-based mapping conversions are considered as follows:

$$\begin{aligned} F^{\alpha}_{\text{sum}}(f_i) &= \sum_{c=1}^{\text{C}} \left| f_i^c \right|^{\alpha} \\ F^{\alpha}_{\text{max}}(f_i) &= \max_{i=1,C} \left| f_i^c \right|^{\alpha}. \end{aligned} \tag{11}$$

Then we calculate the spatial attention weight of the teacher and student instance and obtain the difference between them as the feature enhancement weight

$$\omega_i^f = \left| F\left(S_i^f\right) - F\left(T_i^f\right) \right|. \tag{12}$$

The bigger the ω_i^f, the more different feature knowledge between i^t and i^s. Therefore, ω_i^f will enhance the feature learning between two models. For a better description,

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we present the knowledge flow from the student to the teacher as below

$$\mathcal{L}_{\text{fea}} = \sum_{i \in I} \omega_i^f \left\| S_i^f - T_i^f \right\|. \tag{13}$$

Note that the knowledge flow from the teacher to the student is the same as (13), but the gradients of the teacher model are updated.

2) Category Balance: Category imbalance usually confuses detector training and urges the model to detect instances with more objects of the category. But the accuracy evaluation includes all the classes, which makes the detector need more robustness for all the classes. To settle the issue, we propose a strategy named category balance to facilitate the different category objects to get equal attention and balance category knowledge learning.

Let the instance set I have the ground truth set $G = \{g_1, g_2, \ldots, g_p\}$. For the instance i_n, we define g_i as its corresponding ground truth. The category balance coefficient is given as follows:

$$\omega_i^y = 1 - \frac{\psi(g_i)}{|G|} \varphi(y_i) \tag{14}$$

where $|G|$ represents the number of ground truth in G. $\psi(g_i)$ denotes the number of ground truth for the same category as g_i. y_i is the classification prediction of instance i and $\varphi(y_i)$ signifies the max probability of all the category predictions. Equation (14) can be interpreted as below. When $\varphi(y_i)$ is fixed, the higher the number of ground truth belonging to a category, the smaller the category balance coefficient. Thus, it suppresses the gradient optimization of the larger categories and increases the weights of the smaller categories. Easy instance usually has a higher $\varphi(y_i)$ and hard instance usually has lower $\varphi(y_i)$. When $\psi(g_i)/|G|$ is fixed, ω_i^y will give the easy instance less weight and endue the hard instance more weight. Therefore, ω_i^y has a category balance impact for each sample, and category learning is defined as follows:

$$\mathcal{L}_{\text{cate}} = \sum_{i \in I} \omega_i^y \left\| S_i^y / \text{tmp} - T_i^y / \text{tmp} \right\| \tag{15}$$

where tmp represents the temperature coefficient hyperparameter to adjust the distribution.

3) Location Calibration: Given $\{u_1, v_1, u_2, v_2\}$ is the distance of four directions based on the location of the instance. We calculate the location punishment coefficient σ for this instance as below

$$\sigma = \frac{|u_1 - u_2|}{2(u_1 + u_2)} + \frac{|v_1 - v_2|}{2(v_1 + v_2)} \tag{16}$$

where $\sigma \in [0, 1]$. The closer the instance box to the corresponding ground truth, the less the σ. Then we compute the location punishment coefficient of the student model and the teacher model and get the location calibration weight as follows:

$$\omega_i^b = \left| \sigma\left(S_i^b\right) - \sigma\left(T_i^b\right) \right|. \tag{17}$$

The bigger the ω_i^b, the more difference in location accuracy between the student and the teacher. We adopt ω_i^b as the location calibration weight to enhance the location knowledge relationship between the student and the teacher. The location learning is defined as follows:

$$\mathcal{L}_{\text{loc}} = \sum_{i \in I} \omega_i^b \text{GIoU}\left(S_i^b, T_i^b\right) \tag{18}$$

where GIoU means that knowledge transfer is implemented by GIoU loss.

We summarize the above IPP strategy (see Fig. 6) and apply them to feature, category, and location knowledge transfer, respectively. The whole distillation optimization loss is presented below

$$\mathcal{L}_{\text{distillation}} = \mathcal{L}_{\text{fea}} + \mathcal{L}_{\text{cate}} + \mathcal{L}_{\text{loc}}. \tag{19}$$

D. Optimization

In this part, we present the optimization for the DIL framework, which adopts a unique distillation strategy. The teacher and student models are trained together, and both of their parameters are updated during the training phase. We first present the student model loss function as below

$$\mathcal{L}_s = \mathcal{L}^s_{\text{detection}} + \mathcal{L}^s_{\text{distillation}} \tag{20}$$

where $\mathcal{L}^s_{\text{detection}}$ is the object detection loss, and $\mathcal{L}^s_{\text{distillation}}$ is the distillation loss.

The teacher model optimization is similar to the student model, and the loss function is defined as follows:

$$\mathcal{L}_t = \mathcal{L}^t_{\text{detection}} + \mathcal{L}^t_{\text{distillation}}. \tag{21}$$

The joint training between the student model and the teacher model makes sure the gaps in data and parameters are small enough, which is better for lightweight model optimization. The pipeline of the DIL is elucidated in Algorithm 1.

IV. Experiment

In this part, the experimental details that contain datasets and implementation settings are introduced first. Then we will conduct the ablation experiments and analyze the results. Finally, we compare the proposed DIL with other SOTA distillation approaches. Moreover, to verify the performance of our lightweight model, we also compare it with other detectors.

A. Dataset

1) DOTA: Xia et al. [34] propose the DOTA benchmark for RS object detection task in 2018. We mainly conduct experiments on the horizontal bounding box HBB detection task. There are 2806 large images and 188 282 instances in the dataset, and 15 categories are involved. The size of the original varies from 800 × 800 to 4000 × 4000. Images are cropped into 1024 × 1024 pixels with an overlap of 200 pixels. 1/2 images are selected for training, 1/6 are chosen for validation, and 1/3 are for testing.

2) NWPU VHR-10: The second dataset is NWPU VHR-10 which is proposed by Cheng et al. [35]. It includes 800 images, and 650 images of them have objects. We adopt 520 images as the training set and 130 images as the testing set. Ten categories are divided in the dataset and we resize the image to 1000 × 600 in the experiments.

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Algorithm 1 DIL

Input:
$\mathcal{X}$ are images and $\mathcal{Y}$ are labels;
$\mathcal{D} = \{\mathcal{X}, \mathcal{Y}\}$ is the training set;
T, θ_t are the teacher model and its parameters;
Output:
S, θ_s are the student model and its parameters;

Load the weights of T and initialize the parameters of S
for mini-batch $\mathcal{D}_b = \{\mathcal{X}_b, \mathcal{Y}_b\}$ **do**
  $I_t \leftarrow T(\mathcal{X}_b, \theta_t)$, $I_s \leftarrow S(\mathcal{X}_b, \theta_s)$
  for $i \in I$ **do**
    $P_{i^t} \leftarrow T(i^t, \theta_t)$, $P_{i^s} \leftarrow S(i^s, \theta_s)$
    $\{\omega_{i^t}, \omega_{i^s}\} \leftarrow \{P_{i^t}, P_{i^s}\}$
    $V(i^t) = 1/\mathcal{L}(i^t, \mathcal{Y}_b^{i^t})$, $V(i^s) = 1/\mathcal{L}(i^s, \mathcal{Y}_b^{i^s})$
    if $V(i^t) < V(i^s)$ **then**
      $S \Leftarrow^{\omega_i} T\ (\mathcal{L}^t_{distillaltion})$
    else $\{V(i^t) > V(i^s)\}$
      $T \Leftarrow^{\omega_i} S\ (\mathcal{L}^s_{distillaltion})$
    end if
  end for
  $\mathcal{L}^t_{detection} = \mathcal{L}(i^t, \mathcal{Y}_b^{i_t})$
  $\mathcal{L}^s_{detection} = \mathcal{L}(i^s, \mathcal{Y}_b^{i_s})$
  $\mathcal{L}_s = \mathcal{L}^s_{distillation} + \mathcal{L}^s_{detection}$
  $\mathcal{L}_t = \mathcal{L}^t_{distillation} + \mathcal{L}^t_{detection}$
  $S : \theta_s = argmin_{\theta_s}\mathcal{L}_s$;
  $T : \theta_t = argmin_{\theta_t}\mathcal{L}_t$;
end for
return $\mathcal{S}, \theta_s$

3) DIOR: Li et al. [36] propose the DIOR benchmark for the RS object detection task in 2020. About 23 463 images and 192 472 instances are in the datasets. We adopt 800 × 800 as the image size during the experiments. In general, 11 725 images are selected for training and 11 738 images are chosen for testing.

B. Implementation Details

1) Model Setup: We adopt the classic one-stage detector as the teacher and student models. Moreover, the detector usually contains the backbone, feature pyramid network (FPN), and detection heads. Details are presented as follows.

Teacher: We adopt **ResNet101** as the teacher backbone on the DIOR and DOTA datasets. We use **ResNet50** as the teacher backbone in the small NWPU VHR-10 benchmark.

Student: The student is a lightweight model which adopts **ResNet18** as the backbone. Moreover, we reduce the channel numbers of the decoder and FPN output layers to half for the DIOR and DOTA benchmarks. As for the NWPU VHR-10 benchmark, the channels are reduced to a quarter of the original layers.

2) Training Strategy: Different training strategies are used for diverse datasets, which are presented in Table I. The optimizer adopts the stochastic gradient descent (SGD) algorithm. We conduct 500 rounds of warm-up with a 0.001 learning rate. The momentum and weight decay are set as 0.9 and 0.0001, respectively. Besides, we use the model pretrained in ImageNet for backbone initialization for faster model convergence. In the test phase, the nonmaximum suppression IoU threshold is 0.4 for DIOR and DOTA and 0.6 for NWPU VHR-10. An Nvidia 3090 GPU is used for conducting experiments.

TABLE I
SETTING

Benchmark	Image Size	Batch Size	Lr	Epoch
NWPU VHR-10	1000×600	4	0.001	40 (30)
DIOR	800×800	8	0.005	24 (20,22)
DOTA	1024×1024	8	0.005	24 (20,22)

3) Evaluation Metric: We adopt the mean average precision (**mAP**) to measure the performance of model on the dataset. Before introducing AP, we briefly explain its relationship with true positive (TP), false positive (FP), false negative (FN), precision and recall (PR). TP means the overlap ratio between the predicted bounding box and the ground-truth box exceeds a threshold. Otherwise, we regard it as an FP. FN will be labeled when a ground truth has no matching prediction results. PR are defined as

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{22}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{23}$$

We get different precisions and recalls by sorting the predicted results according to confidence. Then we draw them into a PR curve and AP is the area below the PR curve. Finally, mAP is got by calculating the average value of all categories' APs.

As for efficiency and lightweight performance of the model, we use FLOPs, FPS, and parameters as indicators.

FPS: We test the speed of the model with "frames per second."

FLOPs: It means the floating point of operations during forward inference.

Parameters (Params): It represents the number of model parameters.

C. Ablation Studies

In this section, we conduct ablation studies on the DIOR dataset.

1) Effectiveness of Dynamic Interactive Learning: We first show the effectiveness of the proposed two modules, which are presented in Table II. Baseline represents the student model which is only trained with ground truth. IIL represents our instance interaction learning, which optimizes the student model by teacher model predictions and ground truth. When the student model is trained with the IIL module, it achieves 68.9% mAP and outperforms the baseline of 3.7% mAP. In addition, we added the IPP module to the base of the ILL module, and three property perception strategies have improved the detector performance by 5.2%.

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TABLE II

ABLATION STUDY OF EACH PART IN DIL. IIL DENOTES INSTANCE INTERACTION LEARNING. IPP REPRESENTS INSTANCE PROPERTY PERCEPTION

Method	IIL	IPP	mAP	ΔAP50
baseline	-	-	65.2	-
DIL	√	-	68.9	+3.7
	√	√	70.4	+5.2

TABLE III

ABLATION STUDY OF INSTANCE INTERACTION LEARNING. FUL DENOTES FROZEN UNIDIRECTIONAL LEARNING, UUL REPRESENTS UPDATED UNIDIRECTIONAL LEARNING, UBL DENOTES UPDATED BIDIRECTIONAL LEARNING, AND IIL IS OUR INSTANCE INTERACTION LEARNING. FOUR STRATEGIES ARE ILLUSTRATED IN FIG. 5

Method	mAP	ΔAP50
baseline	65.2	-
FUL	67.6	+2.2
UUL	68.3	+3.1
UBL	68.5	+3.3
IIL	**68.9**	**+3.7**

2) Effectiveness of Instance Interaction Learning: In Section III-C, we have introduced four distillation strategies. In this part, we will explore four strategies of how to affect knowledge delivery, and the results are presented in Table III. Four strategies including FUL, UUL, UBL, and IIL are illustrated in Fig. 5(a)–(d), respectively. FUL is the classic KD strategy used by most approaches. In FUL, the student is updated and the teacher model is frozen. Besides, the knowledge is the unidirectional transfer to the student from the teacher. UUL is the variant of FUL, and both the student and the teacher are updated during the training phase. In addition, the UBL has broken the traditional teaching mode, and the role of the teacher and the student is determined by their model state. It means that instances in a mini-batch from a model are all the teacher roles or student roles during a learning round. The proposed IIL is the evolution of the UBL and we achieve instance-level interactive learning, which is not restricted by a mini-batch. Each instance has independent distillation and knowledge flow is dynamic between the teacher model and the student model. Instances learn from each other only according to instance value comparison but do not consider which model they are from. On one hand, IIL endows knowledge delivery with more freedom; on the other hand, it keeps the training synchronization training between two models. As shown in Table III, IIL has achieved the SOTA performance compared with other strategies, which reaches 68.9% mAP.

3) Effectiveness of Instance Property Perception: The existing KD in object detection tasks usually ignores the instance property and endows equal weight on the knowledge delivery. However, this strategy is not suitable for instance-level knowledge transfer. In general, instances are diverse in RS images and it is necessary to weigh the knowledge according to instance property. Our IPP module contains three strategies. First, the feature enhancement strategy adopts spatial attention to help the student shrink the feature difference with the teacher model, and hence, the student model can better extract features. Then, the category balance strategy guides the student's instance of harder and fewer category objects to learn key knowledge from the teacher model. Finally, the location calibration strategy facilitates the student to imitate the high-quality bounding boxes of the teacher model. As shown in Table IV, we conduct several perception strategy combination experiments to demonstrate the effectiveness of the IPP module. The alone use of any one of the strategies can effectively improve the detector performance. Among them, FE improves the most accuracy, which achieves 69.6% mAP. The combination of any two strategies also brings a slight boost compared with individual ones. Finally, we add all the three strategies and reach the optimal results of 70.4% mAP.

TABLE IV

ABLATION STUDY OF IPP. FE DENOTES FEATURE ENHANCEMENT, CB REPRESENTS CATEGORY BALANCE, AND LC IS LOCATION CALIBRATION STRATEGY

Method	FE	CB	LC	mAP	ΔAP50
baseline	-	-	-	65.2	-
IPP	√	-	-	69.6	+4.4
	-	√	-	68.5	+3.3
	-	-	√	65.7	+0.5
	√	√	-	69.8	+4.6
	√	-	√	67.7	+2.5
	-	√	√	69.7	+4.5
	√	√	√	**70.4**	**+5.2**

In Section III-C, we propose the feature enhancement strategy which contains several spatial attention conversion. We conduct experiments on the different functions according to (11) and present results in Table V. All the spatial attention conversion functions improve performance. Compared with the first power, the second power reduces the accuracy and it may be owing to that the second power brings optimization more difficult. Moreover, the sum-based function is better than the max-based. In conclusion, F_{sum} achieves the best accuracy and we will use F_{sum} as a spatial attention conversion function.

During classification distillation, the temperature hyperparameter has the role to adjust the probability distribution, thereby providing a soft classification prediction. To find an optimal temperature, we test tmp from 0.25 to 2. As shown in Fig. 7, tmp $= 1$ reaches the best mAP. Hence, we adopt tmp $= 1$ for simplicity.

4) Analysis of Lightweight Performance: In this part, we analyze the lightweight performance of models. As shown in Fig. 8, the abscissa is the inference speed, the ordinate represents the accuracy, and the area of the circle denotes the model parameters. The teacher model has higher accuracy compared with the baseline but with bigger parameters and slower speed. The proposed DIL has validly improved the

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TABLE V
ABLATION STUDY OF SPATIAL ATTENTION FUNCTION

Method	mAP	ΔAP50
baseline	65.2	-
F_{sum}	**69.6**	**+4.4**
F^2_{sum}	67.9	+2.7
F_{max}	68.7	+3.5
F^2_{max}	67.8	+2.6

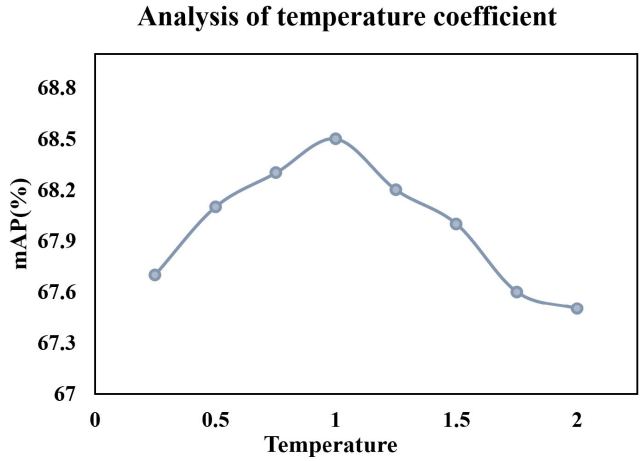


Fig. 7. Illustration of the temperature coefficient influence. During category knowledge learning, we adopt the temperature coefficient to soften the classification outputs.

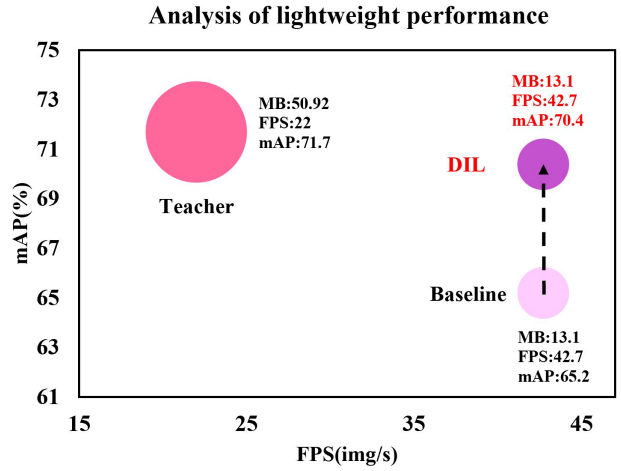


Fig. 8. Illustration of lightweight performance analysis. The area of the circle represents the parameters of the model.

lightweight student model performance without extra overhead. The parameters and inference speed are the same as the original baseline. In conclusion, the results demonstrate the effectiveness of KD and verify the parameter redundancy of the heavy model.

5) Heterogeneous Network Backbones: To prove the universality of our method, we conduct experiments that use heterogeneous student backbones. We adopt MobileNetV2 [1], ShuffleNet [2], and RegNetX-800m [40] to replace the original ResNet18 [19], but the teacher backbone is still

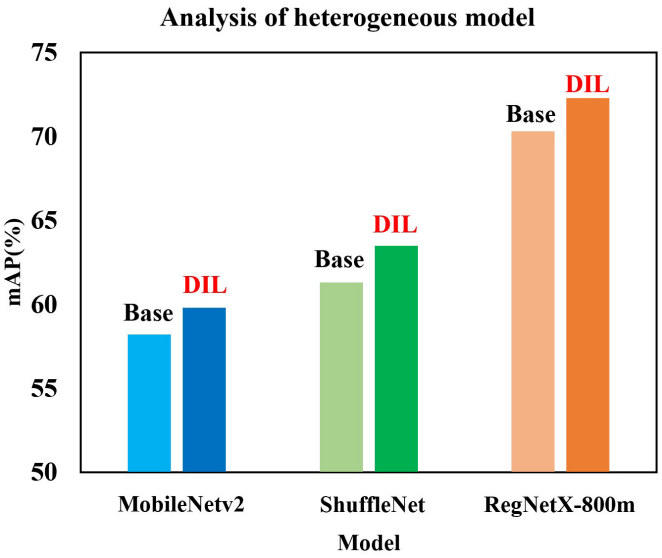


Fig. 9. Illustration of heterogeneous student model analysis. In this experiment, we keep the teacher model consistent but use different structures for the student model.

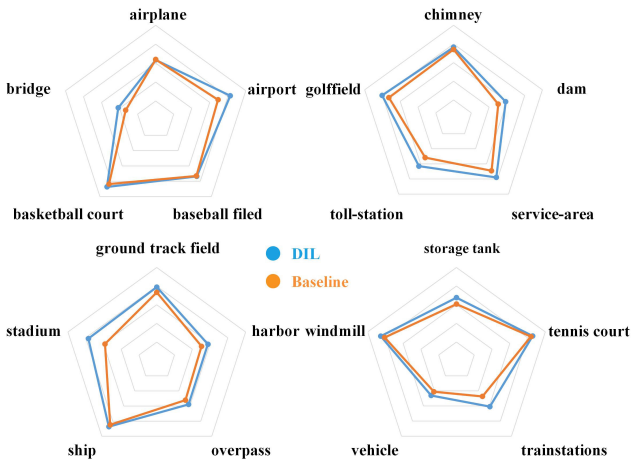


Fig. 10. Illustration of quantitative analysis. We show the AP of 20 categories between the DIL and baseline.

ResNet101 [19]. Three lightweight backbones adopt different network structures compared with the teacher backbone, which means they have different feature extractability. As shown in Fig. 9, when the three heterogeneous backbones are trained with our DIL, their performance has different degrees of improvement. The results also prove the effectiveness of our methods and reveal that we can optimize different lightweight models with a well-trained teacher model.

6) Qualitative Analysis: Fig. 10 presents the comparisons of each category's AP between DIL and baseline. The radar map shows our DIL can effectively improve detection for the category of the small amounts of ground truth including the airport, bridge, dam, service area, toll station, golf field, stadium, and train station. Then we show the attention of feature maps between baseline and DIL, which is illustrated in Fig. 11. The first row is the original image, the second one is the baseline, and the third is the proposed DIL. We select three images including diverse size objects, such as harbors,

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TABLE VI
COMPARISON WITH SOTA DISTILLATION METHODS ON DOTA, NWPU VHR-10, AND DIOR BENCHMARKS

	DOTA		NWPU VHR-10		DIOR	
method	mAP	ΔAP50	mAP	ΔAP50	mAP	ΔAP50
teacher	71.65	-	93.21	-	71.7	-
student	63.80	-	86.56	-	65.2	-
DeFeat [29]	64.11	+0.31	86.92	+0.36	67.7	+2.5
KD [24]	65.80	+2.00	86.82	+0.26	67.2	+2.0
FGFI [5]	66.25	+2.45	88.69	+2.13	69.5	+4.3
FitNet [25]	66.30	+2.50	86.85	+0.29	69.2	+4.0
RKD [37]	67.19	+3.39	90.20	+3.64	67.3	+2.1
TAR [6]	68.09	+4.29	90.22	+3.66	69.5	+4.3
ARSD [31]	68.28	+4.48	90.92	+4.36	70.1	+4.9
S^3MKM [32]	69.00	+5.20	92.90	+6.34	70.2	+5.0
DIL (ours)	**69.20**	**+5.40**	**93.50**	**+6.94**	**70.4**	**+5.2**

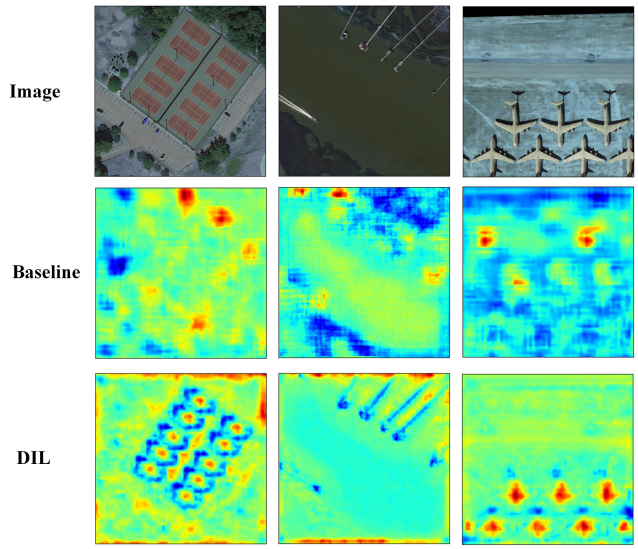


Fig. 11. Visualization of feature map attention between baseline and DIL.

ships, vehicles, airplanes, and tennis courts. The baseline shows worse feature extraction ability, which focuses more on backgrounds. However, DIL pays more attention to objects of diverse sizes, and this denotes that the lightweight detector trained with DIL can better extract core features. As shown in Fig. 12, DIL has more satisfactory detection results compared with baseline, which demonstrates the effectiveness of KD.

D. Comparisons With SOTA Distillation Methods

In this part, the proposed DIL is compared with other SOTA distillation approaches. The same teacher and student model settings are used in experiments, which are presented in Section IV-B1. As shown in Table VI, the student trained with DIL surpasses other distillation methods on three benchmarks.

We first analyze three classic distillations in the classification task. KD [24] provides the soft classification prediction distributions of the teacher model, which generates the extra soft label for student learning. The soft label enhances the learning of shared features of similar categories by the student models, which improves 2%/0.26%/2.5% performance. FitNet [25] focuses more on feature knowledge, which promotes the student model imitating the teacher's feature map. The feature imitations bring the promotion of 2.50%/0.29%/4.0%. Relational knowledge distillation (RKD) [37] puts forward relationship KD to mine the relationship among samples. The relationship among instances transmits the structure information, which represents the implicit knowledge, and it improves 3.39%/3.64%/2.1% accuracy finally. Then, we analyze three distillation approaches designed for object detection in the natural scene. FGFI [5] adopts the features close to the object region rather than all the feature maps as knowledge, which validly reduces the impact of the background, and the results present that the special design is more beneficial to object detection. TAR [6] proposes a 2-D Gauss mask to obtain ground-truth information and uses the classification and regression results of positive samples as extra knowledge. Multi-information distillation is more fit to the multitask character of detection, which improves 4.29%/3.66%/4.3% performance. While Defeat [29] suggests a certain extent background knowledge also benefits performance promotion. But owing to the huge and complex background in the RS image, it seems to not work well on RS benchmarks. Finally, we will introduce two designed for RS scene distillation frameworks, ARSD [31] and S^3MKM [32]. They proposed that multiscale feature distillation and statistical sample selection are better for RS instances distillation. They promote 4.48%/4.36%4.9% and 5.40%6.34%5.2% accuracy, respectively.

However, all the above approaches are offline unidirectional distillation. The offline training strategy leads to asynchronization updates between the teacher model and the student model. The unidirectional distillation results in the teacher model may misguide the student in some knowledge. Besides, they do not deliver dynamic knowledge according to instance property but treat different instances equally. There are two dynamic thoughts in the proposed DIL. One is that instances

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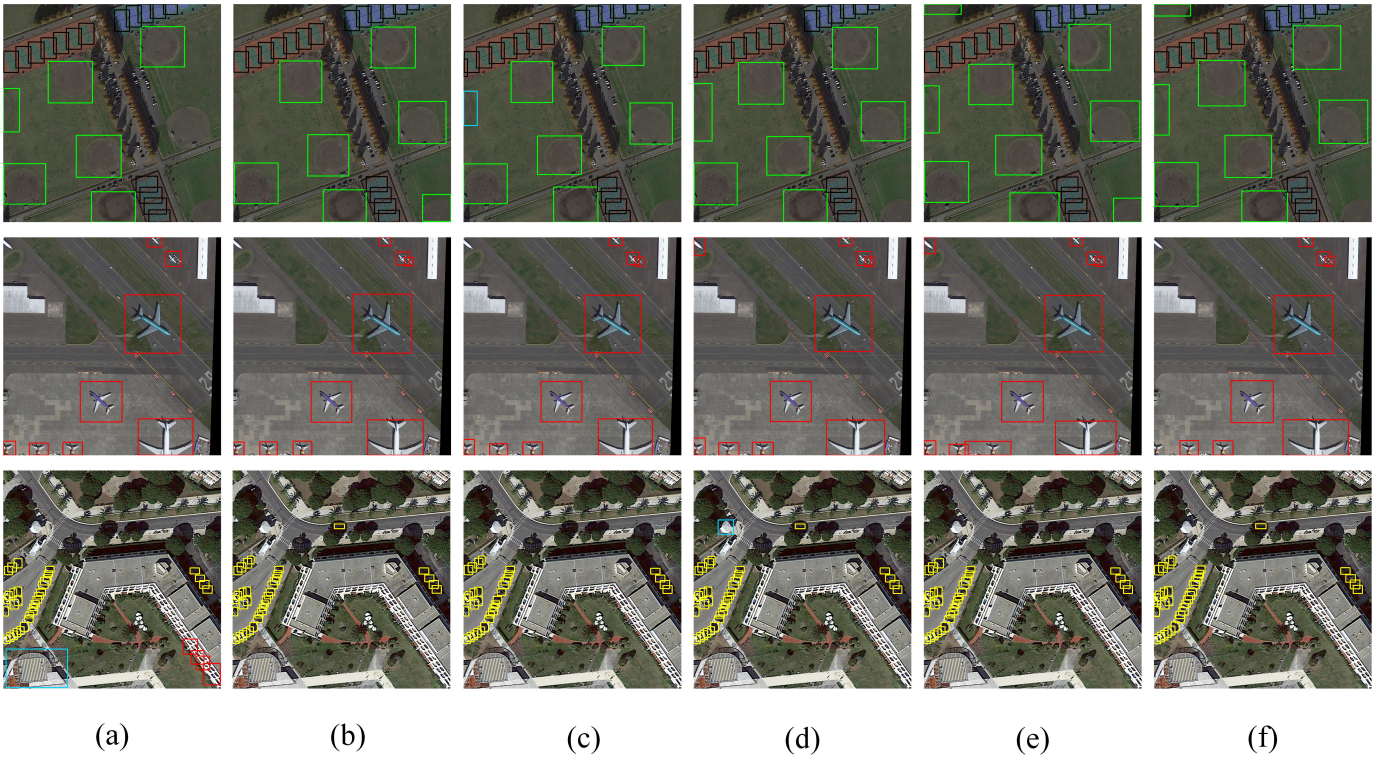


Fig. 12. Comparison of visualization results among diverse distillation approaches. (a) Baseline. (b) DIL. (c) KD. (d) FitNet. (e) FGFI. (f) TAR. The categories of objects in the figure correspond to the colors as follows. "Airplane" is red, "Baseball" is green, "Tennis court" is black, "Vehicle" is yellow, and "Storage tank" is blue.

TABLE VII
COMPARISON WITH CLASSIC DETECTORS ON DOTA, NWPU VHR-10, AND DIOR BENCHMARKS

Approach	DOTA				NWPU VHR-10				DIOR			
	mAP (%)	FPS (f/s)	Params (M)	FLOPs (G)	mAP (%)	FPS (f/s)	Params (M)	FLOPs (G)	mAP (%)	FPS (f/s)	Params (M)	FLOPs (G)
Faster R-CNN [7]	60.64	7.0	60.19	289.25	77.80	25.4	41.17	127.70	54.1	16.8	60.21	182.02
YOLOv3 [10]	60.00	13.0	61.63	198.92	88.30	31.2	61.57	121.27	57.1	14.2	61.95	122.22
RetinaNet [12]	50.39	14.0	55.39	293.36	89.40	26.3	36.29	123.27	65.7	18.2	55.49	180.62
ATSS [13]	66.84	26.6	18.97	156.01	90.50	37.4	18.96	89.90	67.7	26.7	18.98	95.50
FCOS [38]	67.72	18.3	31.57	202.15	89.65	24.4	31.86	116.63	67.6	25.4	31.88	123.51
GFL [39]	66.53	23.8	19.13	159.18	88.80	34.7	19.13	91.73	68.0	36.5	19.13	97.43
MobileNetV2 [1]	56.91	22.5	10.30	124.24	76.9	30.2	10.29	71.49	58.2	30.1	10.32	76.1
ShuffleNet [2]	57.73	22.2	12.11	142.60	83.0	25.9	12.10	82.17	61.3	25.0	12.12	87.31
DIL (ours)	**69.20**	**36.7**	**13.08**	**68.03**	**93.50**	**47.1**	**11.57**	**26.65**	**70.4**	**42.7**	**13.10**	**41.60**

can guide each other by state comparison and the other is that knowledge transfer is weighted by instance properties. The two modules make us achieve the SOTA performance (5.40%/6.94%/5.2%) compared with other approaches in three benchmarks.

As shown in Fig. 12, we present some visualization results for the qualitative analysis. Fig. 12(a) is the baseline, Fig. 12(b) is the DIL, and Fig. 12(c)–(f) shows other distillation methods. Compared with baseline and other approaches, DIL has less miss detection and false alarm.

E. Comparison With the Classic Detectors

In this section, the optimized lightweight model is compared with other classic detection algorithms. The comparison includes heavy detectors and lightweight detectors, and the results are presented in Table VII. Compared with the two-stage, single-stage, anchor-based, or anchor-free heavy detectors, the proposed DIL has higher accuracy, smaller parameters, fewer FLOPs, and faster inference speed. Moreover, the comparison results with the lightweight models such as MobileNetV2 [1] and ShuffleNet [2] also show the superiority of our method. Although close parameters are among the

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lightweight models, the accuracy of DIL surpasses the other two models by a wide margin. The use of depth separable convolutions in MobileNetV2 [1] and ShuffleNet [2] reduces the inference speed and increases FLOPs compared with us. In conclusion, the proposed DIL shows the potency of KD, which validly promotes the performance of the detector without extra overhead.

V. Conclusion

In this article, we propose a DIL framework for lightweight RS detector optimization, which includes the instance interaction learning module and IPP module. The proposed IIL dynamically calculates the value of each instance and facilitates the low-quality instance to learn from the high-quality one whether it comes from the teacher model or the student model. The IIL module urges the teacher model and the student model to learn from each other, which keeps the synchronization between the two models and contributes to model training. Moreover, we also propose the IPP module that contains feature enhancement, category balance, and location calibration. The three strategies dynamically weigh the different knowledge according to the instance property, which is better for knowledge transfer between RS instances. To prove the effectiveness of the DIL framework, we conduct extensive experiments on three benchmarks, and the results demonstrate the SOTA performance of our approaches. Furthermore, the lightweight detector trained with the proposed DIL exceeds both heavy and lightweight detectors. In our future work, we will pay more attention to the combination of KD and other lightweight model optimization approaches.

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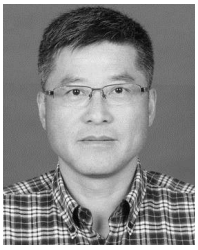
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